



# AUTONOMOUS WAITER

Department of Mechanical Engineering  
Bangladesh University of Engineering & Technology  
Dhaka: 1000

**Course No: ME 362**

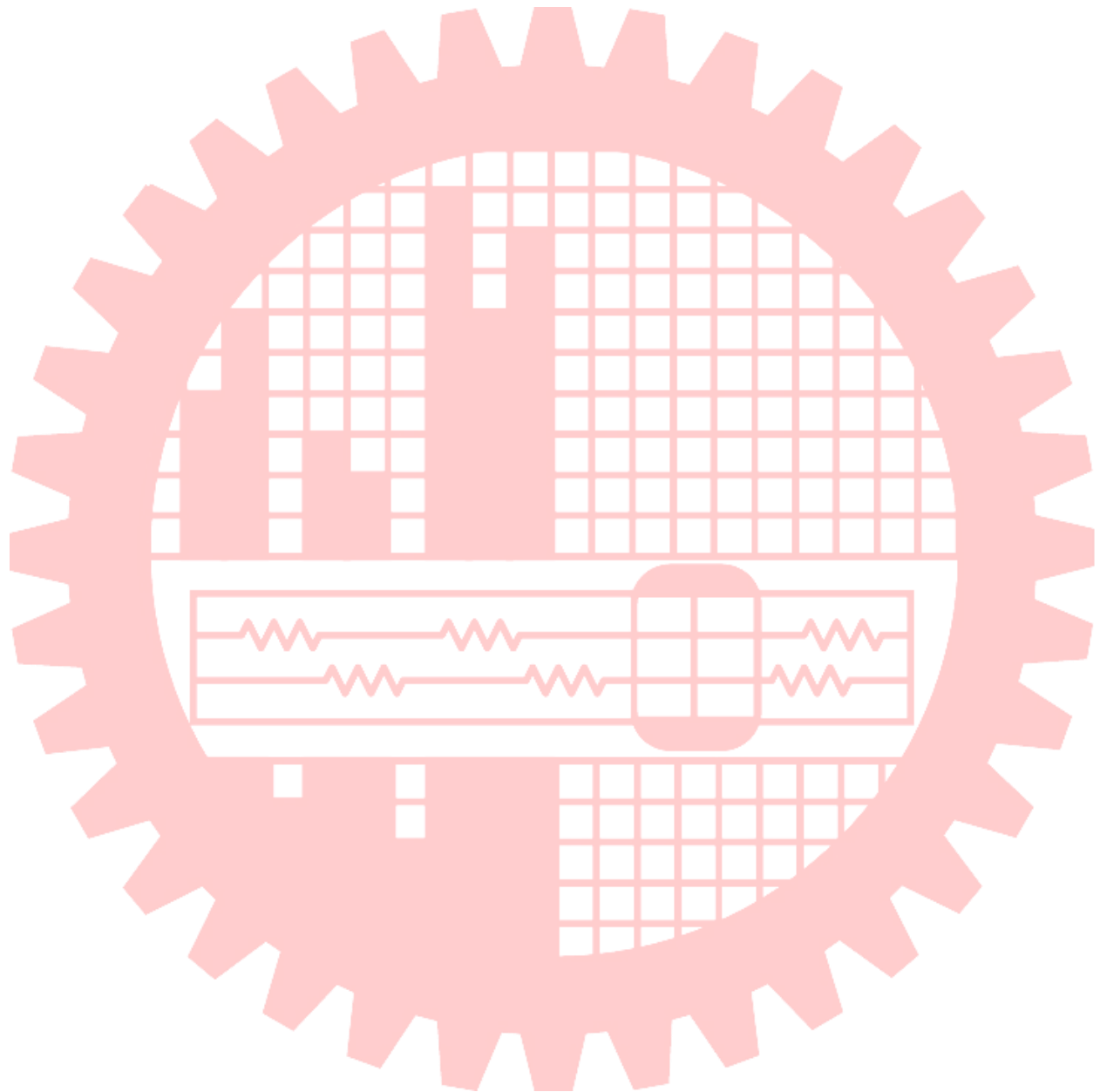
**Course Title: Instrumentation and Measurement Sessional**

**Group No: A28**

## ACCOMPLISHED BY

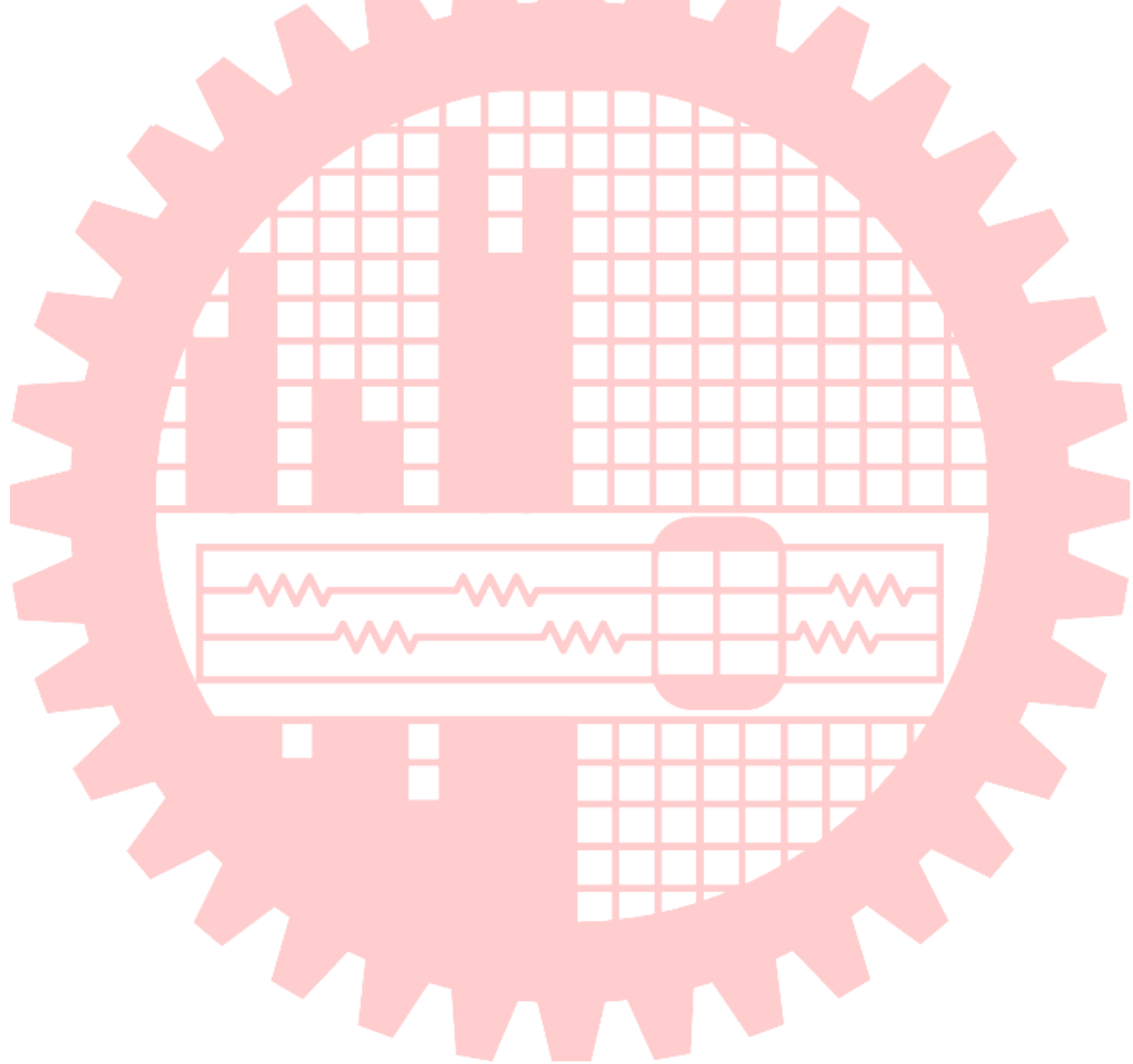
KHALED MOSHARRAF MUKUT (1010046)  
MD.MOHAIMINUL ISLAM KHAN (1010038)  
SAADBIN KHAN (1010054)

Copyright © by Bangladesh University of Engineering and Technology. All rights reserved. The contents of the authors and Bangladesh University of Engineering and Technology. Any part of this document (electrical or mechanical or photocopy) cannot be rebuilt or reproduced without the permission from the authority of Bangladesh University of Engineering and Technology.



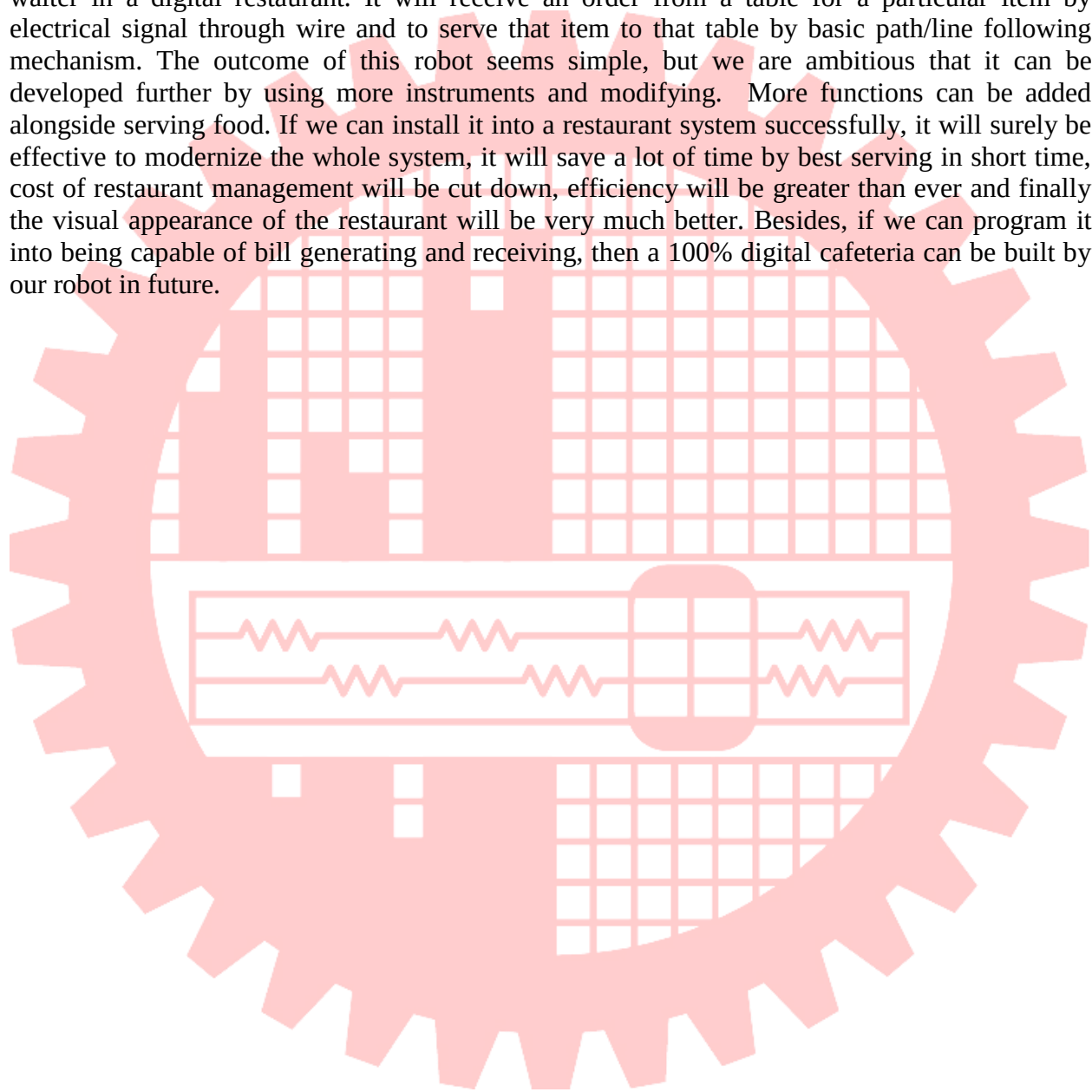
## **Acknowledgement**

First of all we would like to thank our almighty creator for the completion of our first team project as undergraduate students of Bangladesh University of Engineering and Technology. For completing our task we are very much grateful to our honorable teachers. Dr. M. A. Rashid Sarkar, Mr. Kazi Arafat Rahman, Mr. Aminul Islam Khan were all very helpful in and out of classroom to guide us in our successful completion of the project.



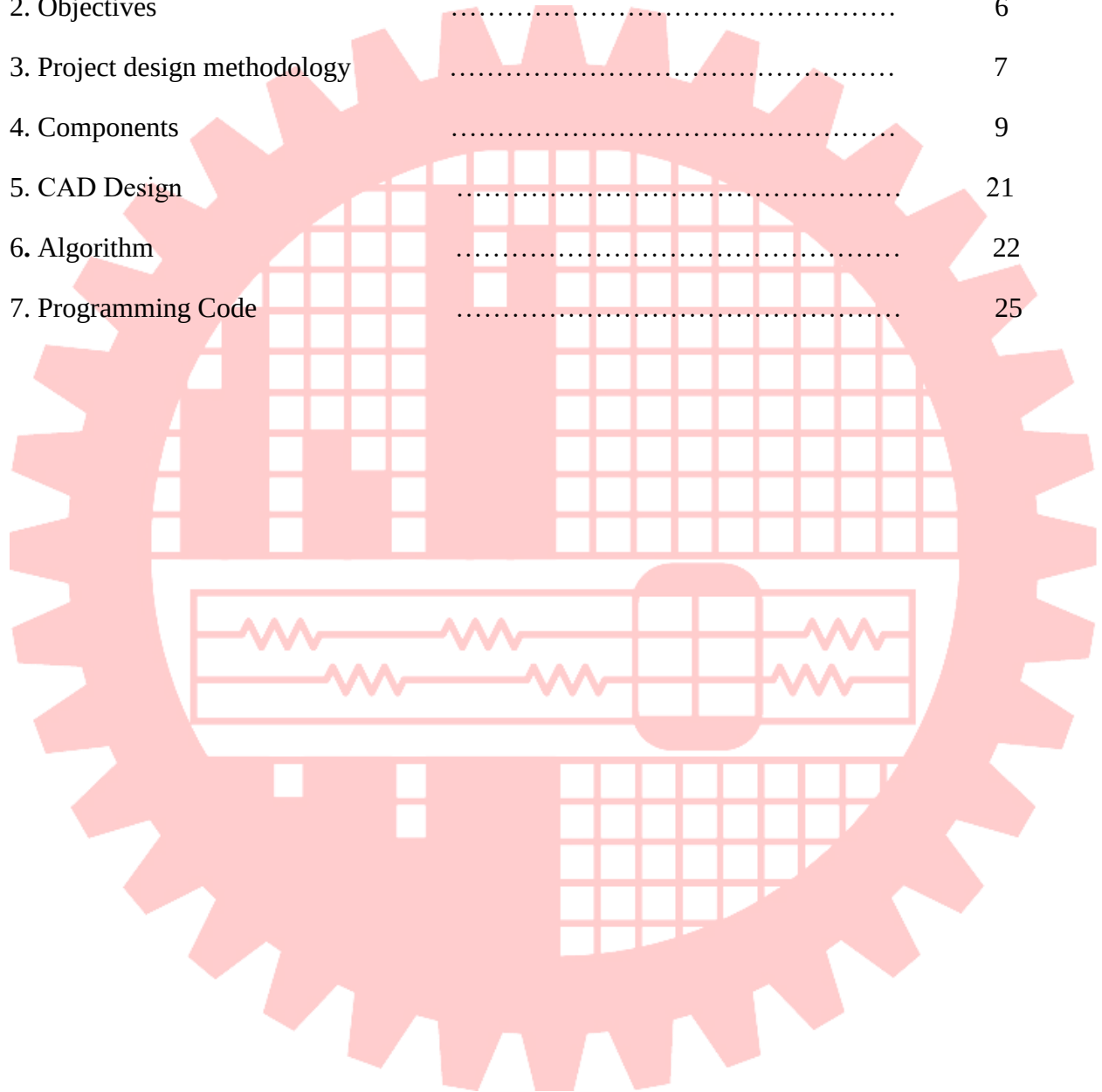
## **Abstract**

Robot is a device that automatically performs complicated and often repetitive tasks, it is built on mechanism guided by automatic controls. It has been invented to reduce the human effort and waste of time in a great range. Our project is to make an autonomous robot which will serve as a waiter in a digital restaurant. It will receive an order from a table for a particular item by electrical signal through wire and to serve that item to that table by basic path/line following mechanism. The outcome of this robot seems simple, but we are ambitious that it can be developed further by using more instruments and modifying. More functions can be added alongside serving food. If we can install it into a restaurant system successfully, it will surely be effective to modernize the whole system, it will save a lot of time by best serving in short time, cost of restaurant management will be cut down, efficiency will be greater than ever and finally the visual appearance of the restaurant will be very much better. Besides, if we can program it into being capable of bill generating and receiving, then a 100% digital cafeteria can be built by our robot in future.



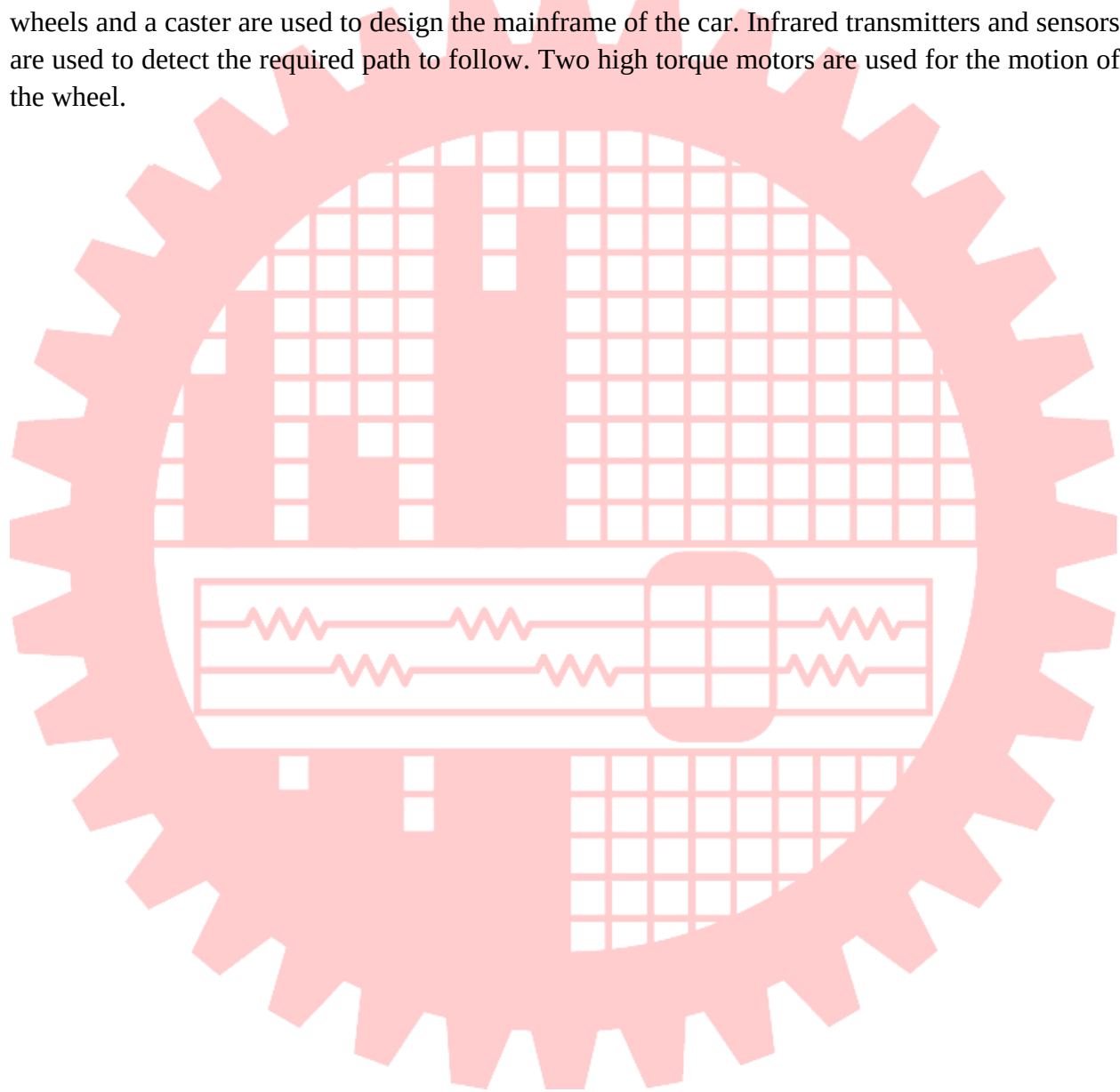
## Table of contents

|                               |       |    |
|-------------------------------|-------|----|
| 1. Introduction               | ..... | 5  |
| 2. Objectives                 | ..... | 6  |
| 3. Project design methodology | ..... | 7  |
| 4. Components                 | ..... | 9  |
| 5. CAD Design                 | ..... | 21 |
| 6. Algorithm                  | ..... | 22 |
| 7. Programming Code           | ..... | 25 |



## **Introduction:**

Our project Autonomous Waiter Robot is an automatically controlled car which will take an order for a particular item from a particular table and serve the item by following line. It will start from the counter, reach the desired table, customer will receive the item, and then it will turn around and return to the counter. Two acrylic boards one on another with distance, two wheels and a caster are used to design the mainframe of the car. Infrared transmitters and sensors are used to detect the required path to follow. Two high torque motors are used for the motion of the wheel.



## **Objective:**

Main objective of Autonomous Waiter Robot is to receive an order from a table and serve the item there, which deducts the need of waiters in the restaurants. If we consider the aspect of Bangladesh, we can at least reduce the problems of the highly messed up restaurant systems of our country. Moreover, by further developing we can build up a whole digital cafeteria, which will be a great step in the technological development of our country, though we have not looked through that far yet in this project. Our goal till now is to make an autonomous device that will just vanish the need of human labor in serving in a restaurant.

## **Difficulties Had to Face:**

- There were so many reflected lights around the car. So, it was difficult to calibrate it to the proper light.
- Synchronization between the table circuit and the car circuit has been a great problem. A wireless system could have been a much better idea in this regard which will be considered in future development of the project.

## **METHODOLOGY:**

### **Tasks Involved:**

The robot has sensors and receivers. So, it can sense the reflected light from the surface it sends the value to the microcontroller where the value is compared with the calibrated value. Using this comparison microcontroller sends command to the motors and the motors turns on or off according to the command. The code used for the project command in such way that the robot will only follow the black line and to avoid the white line. To turn left or right first it has to sense the infrared transmitter that lies beside the road than it will find the black branch of the road. It will then turn to it and follow the line. Again it will sense the end of the road (where the table stands) by white surface and will stop in front of the table. For these sensing purposes we used IR sensors and receivers. There were Infrared transmitters on the road to show the robot which way to go in the next intersection. Every sensor has one of its own transmitter and one receiver. The transmitter transmits light and receiver receives it. That how it distinguish black line from white surface. We used gear motor here. Power is supplied by adaptor or a 12V battery. Here we used ATMEGA 16A as microcontroller which is placed in a bread board. To operate the motor we used an IC L293D.

## **Parts used in the project:**

Components used in this project includes electronics electrical, non-electrical parts. Here we are presenting all the components list by list which were used.

### **Electrical Components**

| Item No. | Component        | Quantity  |
|----------|------------------|-----------|
| 1        | ATMEGA16         | 1         |
| 2        | L293D IC         | 2         |
| 3        | IR sensor        | 6         |
| 4        | IR reciever      | 2         |
| 5        | IR transmitter   | 8         |
| 6        | Potentiometer    | 1         |
| 7        | 7805 IC          | 1         |
| 8        | Resistance(220Ω) | As needed |
| 9        | Breadboard       | 4         |
| 10       | Adapter          | 1         |
| 11       | 12 volt DC motor | 2         |
| 12       | LEDs             | As needed |
| 13       | Capacitors       | As needed |
| 14       | Switches         | As needed |

### **Mechanical components**

| Item no. | Component     | Quantity |
|----------|---------------|----------|
| 1        | Acrylic Base  | 2        |
| 2        | Plastic wheel | 2        |
| 3        | Caster wheel  | 1        |

### **Software Used**

1. Win AVR programmer
2. Proteus professional 7.8
3. Atmel Studio 6.0
4. Extreme Burner

## **Short Description of Components:**

Here is a short description about all the components used in our project.

### **1. Breadboard**

We used 4 different breadboards in this project to construct various circuits we designed in the simulation software. The 4 different circuits are:

- Motor operating circuit.
- Sensor circuit.
- Switch circuit.
- Order display circuit.

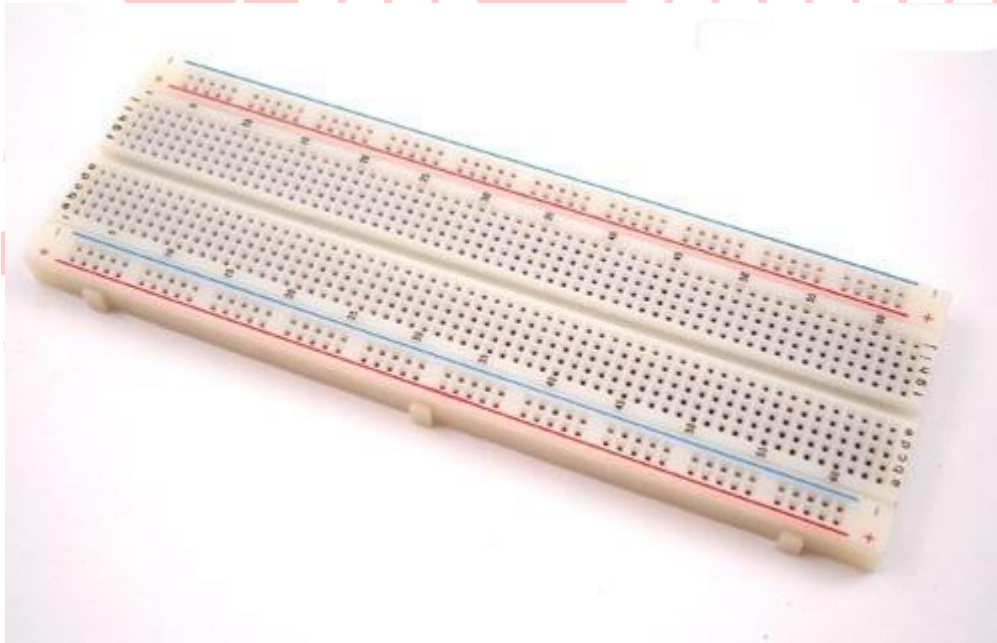


Figure-1: Breadboard

## 2. ATMEGA16

We used a 16 bit microcontroller of ATMEGA series to send the required commands to the motors and sensors.

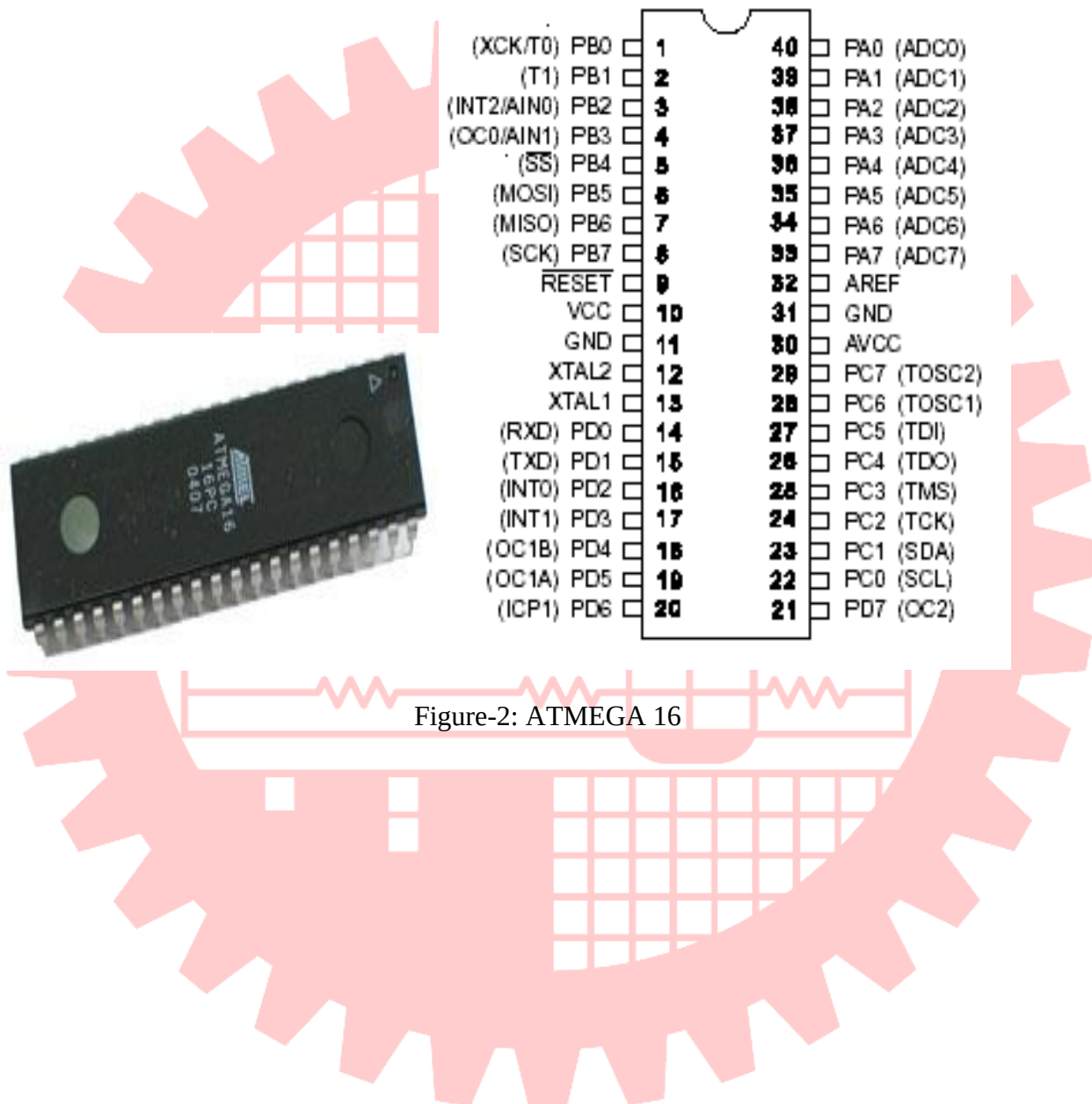


Figure-2: ATMEGA 16

### **3. L293D IC**

L293D is the motor driver IC we used. It is used to control the speed of the motor.

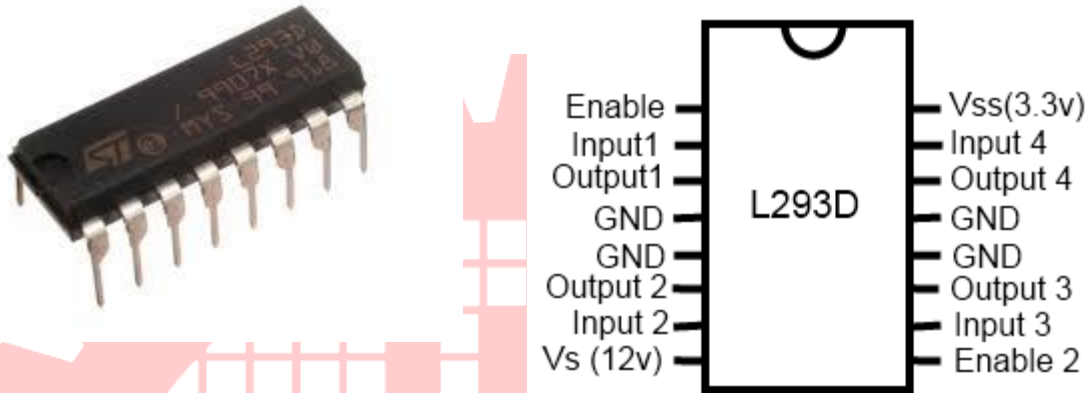


Figure-3: L293D IC

### **4. 7805 IC**

It is the voltage regulator IC we used to supply constant 5 volts to the microcontroller.

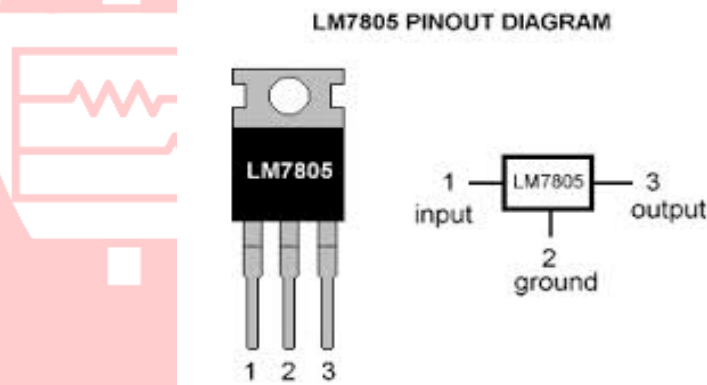


Figure-4: 7805 IC

## **5. Gear Motor**

We used two 12 volt DC motors attached to the wheels of car.

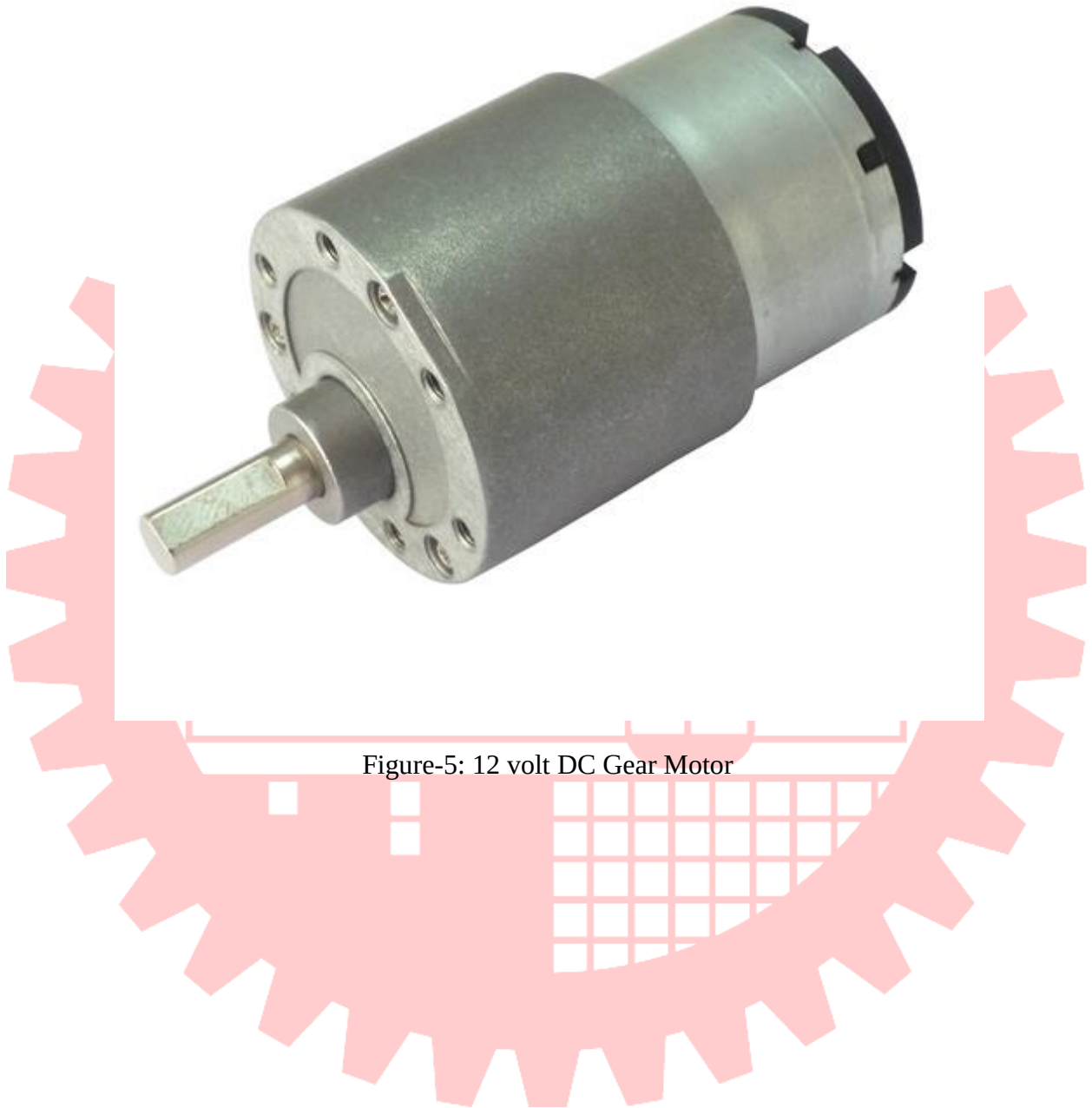
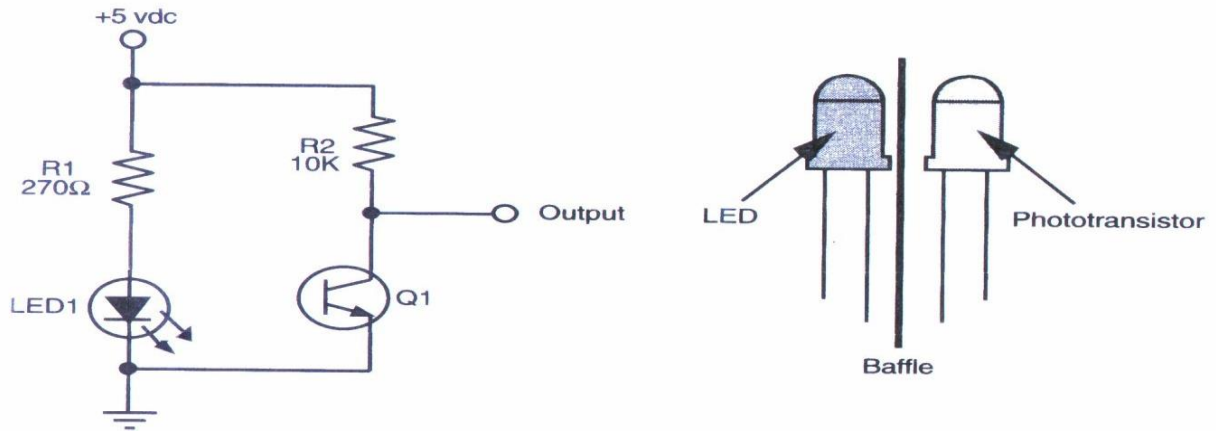


Figure-5: 12 volt DC Gear Motor

## **6. IR Sensor**

We used 5 IR sensors which include Infrared Transmitter and receiver together.



**The basic design of the infrared proximity sensor.**

Figure-6: IR Sensor (Transmitter + Receiver)

## **7. IR Transmitter**

Individual IR transmitters were used to transmit lights from beside the line to point out the path to the table.

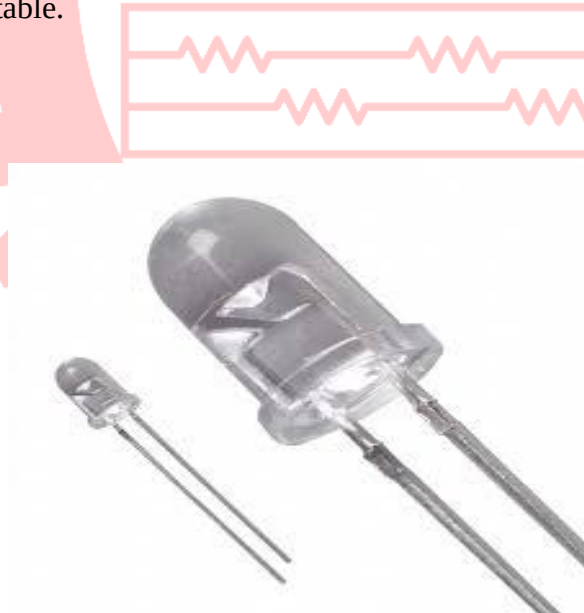


Figure-7: IR Transmitter



Figure-8: IR Receiver

## **8. IR Receiver**

2 IR receivers were set under the car to catch the IR light from the transmitters beside path.

## **9. Potentiometer**

We needed to use variable resistors in several occasions in the project. In those cases we used potentiometers.



Figure-9: Potentiometer

## **10. LED**

We used 9 white LEDs in the order display circuit and 1 LED in motor controlling circuit for checking calibration.



Figure-10: Various LEDs

## **11. Resistances**

We used a lot of resistances throughout the project. Most widely used resistances are 220  $\Omega$ , 1k  $\Omega$ , 10k  $\Omega$ , 2200  $\Omega$  etc.



Figure-11: Resistances

## **12. Switch**

Various types of switches have been used to control current flow throughout the circuits, mainly we used DPDT push-pull switch for our ease.



Figure-12: Switch

### **13. Capacitors**

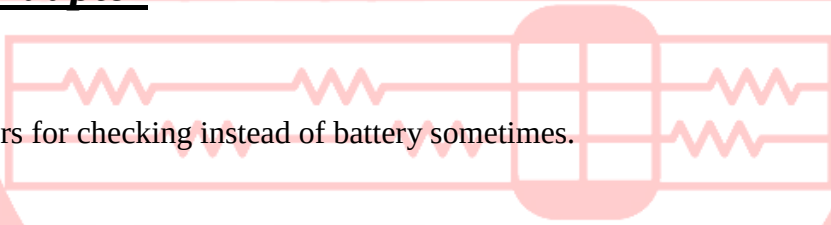
We used capacitors in the circuits in various cases while we needed to reduce noises and for the safety of the ICs too.



Figure-13: Capacitor

### **14. Adapter**

We used adapters for checking instead of battery sometimes.



- **Mechanical Parts**

Mechanical parts we used are described below:

1. **Acrylic Bases**

Two acrylic base plates were used for the mechanical structure of the robot. The sensor circuit was attached under the lower base upside down. Motor controlling circuit was held on the lower base. Upper base is for carrying the food items. In further development we will be able to put a mass sensor on the upper base to make the system more flexible and automated.



Figure-A: Acrylic Bases

## **2. Plastic Wheel**

We used two plastic wheels to run the robot.



Figure-B: Plastic Wheels

## **3. Caster Wheel**

One caster wheel was used at front side of the car to make it easier to revolve.



Figure-C: Caster Wheel

## **Circuit Diagram**

1. Our project has four units. They are:
2. The path.
3. Path detecting Unit.
4. Motor controlling unit.
5. Power supply unit.

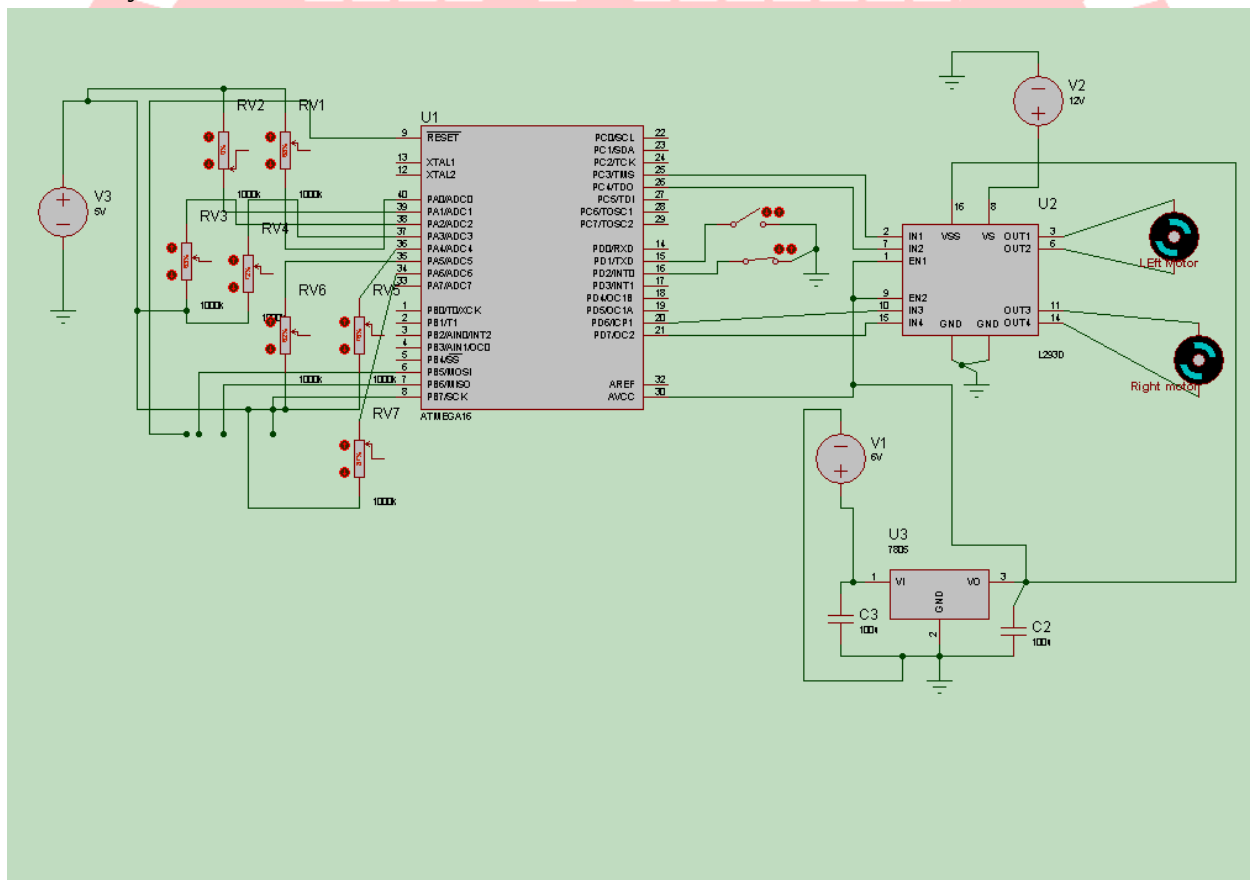
## **The Path**

We used black line on white surface. The robot follows the black lines. There are three different branches of the main line. And so, the main line and the branches make a turn on their connecting point. To show the terminal or the turns IR transmitter are used. There are transmitters at the edge of every turn, so there are three tables on our project path. Every table is at the end of a branch protrudes from the main road/line. At the beginning of the road there is a counter from where the robot starts moving and comes back after delivering to the tables.



## Path Detecting Unit

Our project required a unit to detect the path and keep the Autonomous Robot on the path. For this purpose we made a path sensing circuit which consists of Five IR sensors as receivers and Five IRs as transmitters and also two IR receivers. Two IR sensors are at the front, one directly behind. The IRs at front determines, whether the robot is on the lines and it detects the end of the line. And the one behind it helps the robot following the line. The two receivers are beside the five sensors. One in the left and one in the right. They help to determine the turns and command the branch which branch to follow. The other two sensors includes one at right and one at left. And this two helps the robot to find the black branch beside the white interface. We also used 7 resistors in this circuit. The circuit's IRs are connected to the microcontroller sending signals & the microcontroller uses this feedback to drive the motors accordingly. There is one resistance with every transmitter and receiver.



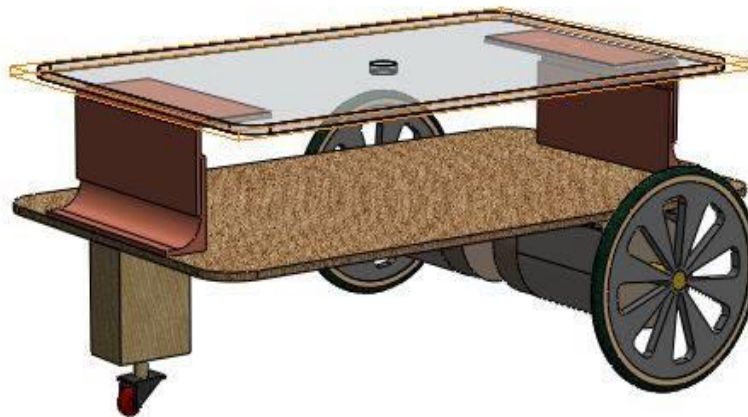
## **Motor Driving Unit**

We used a two wheel mechanism for the motor driving unit. As the motor driving unit we used L293d to drive the rear motors. The circuit's IRs are connected to the microcontroller sending signals to it& the microcontroller is connected to the motor driver and sends signals to the motor driver to drive the motor accordingly. For right turn the right motor is kept off and only the left motor rotates. For left turn vice versa .As the front wheel we used ball caster. As they are more convenient to rotate 360°. We used rubber Pads for the wheel, because the rubbers pads give a better friction between the wheel and the road.

## **Power supply**

During the initial stages we used a 12V-1A adapter to dive our robot. We used it to run our motor and also connected it to a 7805 IC to get 5V output which was used for driving the microcontroller.

## **CAD DESIGN**



## **Algorithm of the project:**

1. Start.
2. Order from one table is placed.
3. Switch on Autonomous waiter.
4. Autonomous waiter will Keep turning right untill find black line with infrared sensor.
5. Sense the line with infrared sensors.
6. Keep moving forward untill find a infrared transmitter transmits light.
7. If (The receiver no.7 senses the transmitter )
  - If (sensor-1 senses a black line) turn left.
  - Find the black line on left and follow it.
  - Else if, keep moving forward.
- 8.Else If (The receiver no.6 senses the transmitter )
  - If (sensor-3 senses a black line) turn right.
  - Find the black line on right and follow it.
  - Else , keep moving forward.
- 9.Else , keep moving forward.
10. Senson 4 & 5 senses white surface and sensor 2 senses black surface.Stop moving.
11. Off the power supply switch.
12. After surving the food – Go to step 3
13. Continue till step 11.
14. End

### **Limitation:**

- We had to use a lot of breadboards instead of PCB which added more weight on the structure.
- We had to make the road with black duct tape, but it is not a perfect absorber, so it could have made some errors in sensor reading.
- We had to make the power supply portable, if sometimes power goes down in the middle of the path, it would be very hard to recover this problem by restoring the system.
- One of the major limitations of our project is that the path has to be preselected.

### **Expected Further Development:**

We have some development ideas that we can use to make the project better, but we couldn't due to shortage of time and resources.

- If we can develop this project further, we would obviously use PCB boards instead of breadboard.
- More sensors could be used to make the run more accurate.
- We designed only simple 3-tables 3 item system, which obviously can be extended resulting into building a digital cafeteria.
- Maze solving code can be added into the program which will solve the limitation of preselecting of paths.

### **Conclusion:**

Considering practical use and the resources we had the project was just a miniature of its real life prototype. Unfortunately the outcome did not met the expectation all the way. It has still a lot to offer. If further work is done developing the project, that can add a new milestone in the run of modern technology.

## **Reference:**

Horowitz, P., Winfield, H., The Art of Electronics, Cambridge University Press, Cambridge

Braga, N. C., Robotics, Mechatronics, and Artificial Intelligence, Newnes, Woburn

## **USEFUL URL's:**

[www.avrfreaks.com](http://www.avrfreaks.com)

[www.en.wikipedia.org](http://www.en.wikipedia.org)

[www.extremeelectronics.co.in](http://www.extremeelectronics.co.in)

[www.micahcarrick.com](http://www.micahcarrick.com)

[www.newbiehack.com](http://www.newbiehack.com)

[www.maxbed.com](http://www.maxbed.com)

[www.ladyada.net](http://www.ladyada.net)

[www.alldatasheet.com](http://www.alldatasheet.com)

[www.howstuffworks.com](http://www.howstuffworks.com)

[www.learnaboutrobots.com](http://www.learnaboutrobots.com)

[www.societyofrobots.com](http://www.societyofrobots.com)

[www.techshop.com](http://www.techshop.com)

# Code for the Microcontroller:

```
#include<avr/io.h>
#include<util/delay.h>

#define fast1 13653
#define fast2 13653
#define medium 6827
#define pause 0

int sensor1, sensor2, sensor3, sensor4, sensor5, sensor6, sensor7;
int threshold[6];
int uv1=0;
int uv2=0;

void calibration()
{
    int cal[7];
    int max[7];
    max[0]=0;
    max[1]=0;
    max[2]=0;
    max[3]=0;
    max[4]=0;
    max[5]=0;
    max[6]=0;

    int min[7];
    min[0]=255;
    min[1]=255;
    min[2]=255;
    min[3]=255;
    min[4]=255;
    min[5]=255;
    min[6]=255;

    int avrg[7];
    avrg[0]=(max[0]+min[0])/2;
    avrg[1]=(max[1]+min[1])/2;
    avrg[2]=(max[2]+min[2])/2;
    avrg[3]=(max[3]+min[3])/2;
    avrg[4]=(max[4]+min[4])/2;
    avrg[5]=(max[5]+min[5])/2;
    avrg[6]=(max[6]+min[6])/2;

    while(bit_is_clear(PIND,1))
    {
        PORTD|=1<<PIND0;
        _delay_ms(200);
        PORTD&=~(1<<PIND0);
        _delay_ms(200);
    }
}
```

```

        for(int i=0;i<7;i++)
        {int threshold[7];
            cal[i]=adc_read(i);

            if(cal[i]<avrg[i])
            {
                min[i]=cal[i];
            }

            elseif(cal[i]>avrg[i])
            {
                max[i]=cal[i];
            }
            avrg[i]=(min[i]+max[i])/2;
            thrghold[i]=avrg[i];
        }
    }
    PORTD&=~(1<<PIND0);
    _delay_ms(200);
}

void forward()
{
    PORTC|=(1<<PC3);
    PORTC&=~(1<<PC4);
    PORTD|=(1<<PD6);
    PORTD&=~(1<<PD7);
}

void left()
{
    PORTC|=(1<<PC3);
    PORTC&=~(1<<PC4);
    PORTD&=~(1<<PD6);
    PORTD&=~(1<<PD7);
}

void right()
{
    PORTC&=~(1<<PC3);
    PORTC&=~(1<<PC4);
    PORTD|=(1<<PD6);
    PORTD&=~(1<<PD7);
}

void stop()
{
    PORTC&=~(1<<PC3);
    PORTC&=~(1<<PC4);
    PORTD&=~(1<<PD6);
    PORTD&=~(1<<PD7);
}

void rightninety()
{
    while(sensor1<threshold[0])
        right();
    stop();
}

```

```

}
void leftninety()
{
    while(sensor3<threshold[0])
        left();
stop();}
void turnabout()
{while(sensor4<threshold[3]&&sensor5<threshold[5])
right();}

int adc_read(int ch)
{
    ch&=0b00000111;////////////////////////
    ADMUX&=0b00000000;
    ADMUX|=ch;
    ADMUX|=1<<REFS0;
    ADCSRA|=1<<ADSC;

    while(!(ADCSRA&(1<<ADIF)));
    ADCSRA|=1<<ADIF;
    return ADC;
}
void adc_init()
{
    ADCSRA|=1<<ADEN|1<<ADPS2;
    ADMUX|=1<<REFS0;
}

int main()
{
    DDRA=0x00;
    DDRD=1<<0|1<<3|1<<4|1<<5|1<<6|1<<7;
    PORTD|=1<<PIND1|1<<PIND2;
    DDRC=0xFF;

    adc_init();
    DDRD|=(1<<PIND4)|(1<<PIND5)|(1<<PIND6)|(1<<PIND7);

    int sensor1,sensor2,sensor3,sensor4,sensor5,sensor6,sensor7;

    while(1)
    {
        sensor1=adc_read(0);
        sensor2=adc_read(1);
        sensor3=adc_read(2);
        sensor4=adc_read(3);
        sensor5=adc_read(4);
        sensor6=adc_read(5);
        sensor7=adc_read(6);
        if(bit_is_clear(PIND,1))

```

```

{
    calibration();
}
if(bit_is_clear(PIND,2))
{
    while(1)
    {
        intp=1;
        _delay_ms(200);
        p=p+1;
        right();
        p=1;
        _delay_ms(200);
        p=p+1;
        left();
        p=1;
        _delay_ms(200);
        p=p+1;
        forward();
        p=1;
        _delay_ms(200);
        p=p+1;
        rightninety();
        p=1;
        _delay_ms(200);
        p=p+1;
        leftninety();
    }
    turnabout();

    if(sensor2>threshold[1]&&sensor4>threshold[3]&&sensor5>threshold[4])
        forward();

    elseif(sensor2>threshold[1]&&sensor4<threshold[3]&&sensor5>threshold[4])
        right();

    elseif(sensor2>threshold[1]&&sensor4>threshold[3]&&sensor5>threshold[4])
        left;

    if(sensor6<threshold[6])
        uv1++;
    if(sensor7<threshold[7])
        uv2++;
    if(sensor3>threshold[2])
    {if(uv1==1)
        rightninety();
      else
        forward();}

    if(sensor1>threshold[0])
    {if(uv2==1)
        leftninety();
      else
        forward();}
    if(sensor4<threshold[3]&&sensor5>threshold[4])
        stop();
    }
    elsestop();
}
}

```